

# Fundamentals of Deep Learning and Programming

## Lecture 0: Course Overview

Fall Semester 2025

Adjunct Professor: Donghoon Kang

---

# Course Overview

**Lecture Time**

**Fridays, 2 PM – 5 PM**

**Classroom**

**Room: L3 301, 302 (3334A, B)**

**Evaluation Criteria**

**Attendance (10%)**

**Programming Assignments (15%)**

**Term-project (15%)**

**Mid-term (30%), Final (30%)**

- Programming assignments: students should submit the **code** and a **2-3 page document** briefly explaining the code to me by **email** ([chocopie@kist.re.kr](mailto:chocopie@kist.re.kr))

# Term Project

Must be performed **alone**

- Should **select a paper** published only in **CVPR, ICCV, ECCV, or NeurIPS after 2023 (inclusive)**.
- Should **write an 8-10 page report** analyzing the training code of the selected paper.  
(including the **experimental results** using **a subset** of training data)

Before Week 7

Should decide which **paper** to **choose**. (also possible to change another paper later)

Before Week 15

Should **submit** an 8-10 page **report** as a term-project.

# Schedule

September

**Week 1**

**Course overview**

**Introduction to Python Programming**

**Week 2**

**Useful Python Libraries**

Numpy, PyTorch

**Week 3**

**Neural Networks Basics**

**Week 4**

**Convolutional Neural Networks (CNNs)**

# Schedule

October

**Week 5**

**Models using CNN**

**Week 6**

**Introduction to Natural Language Processing (NLP)**

**Week 7**

**Attention Mechanism**

**Week 8**

**Transformer**

**Part 1**

**Week 9**

**Mid term exam**

# Schedule

November

**Week 10**

**Transformer**

**Part 2**

**Week 11**

**Transformer**

**Part 3**

**Vision Transformer (ViT)**

**Week 12**

**Generative Models – Introduction to GANs, VAEs**

**Week 13**

**Diffusion Models**

**Part 1**

# Schedule

December

**Week 14**

**Diffusion Models**

**Part 2**

**Week 15**

**Diffusion Models**

**Part 3**

**Week 16**

**Final exam**